BIODATA

Viraj Sanghavi

Computer Engineering Undergraduate | AI/ML and Federated Learning Research

Personal Details

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Education

Degree	B.Tech in Computer Engineering (expected 2028)
Institution	Dwarkadas J. Sanghvi College of Engineering, Mumbai
Current Status	Semester 5 (ongoing), Sept 2024 to present
CGPA	9.15
Coursework	Data Structures, Algorithms, Statistics, Machine Learning

Research

Publicity & Research Member, DJS Anveshan (Research Club), DJSCE Ongoing

Topic: Failure Modes in Federated Learning for Cross-Institutional Fraud Detection

Status: early stage; problem framing and literature review in progress

Problem being studied

- Byzantine-robust aggregators (Krum, FLAME) filter client updates that look unlike the majority. An honest institution with non-IID data can look the same as an attacker, so its valid updates may be discarded.
- The global model can then report high aggregate accuracy while failing on that institution's fraud patterns, with no signal at the server.

Current work

- Refining the problem statement and threat model (e.g. two similar international banks and one regional bank in a joint fraud model).
- Building a literature map of robust aggregation under heterogeneous data to identify what remains open.

Planned next steps: form testable hypotheses, then run federated simulations on public fraud datasets comparing FedAvg, Krum, Median, Trimmed Mean and FLAME using per-client metrics.

Keywords: Federated Learning, Byzantine Robustness, Non-IID Data, Fraud Detection

Research-Oriented Projects

Vigilia, Graph-Native AML Detection Jan 2025 to present

- Started as Argus at iDEA 2.0 (Union Bank of India, Top 40+ finalist); detected 8 structural AML typologies and 7 behavioural signals on 500K+ synthetic transactions; being rebuilt with a GNN detection layer.

TinyML, CIFAR-10 on a Microcontroller Budget Oct 2026 to present

- Compressed a ResNet-18 teacher (93.3%) to an int8 model at 84.8% accuracy in 70.6 KB flash and 16 KB peak RAM via Once-for-All NAS, structured pruning, and KD + QAT, with every stage benchmarked against baselines.

Professional Experience

- **Developer Intern**, Aelius Turbina LLP, Mumbai — Sept 2026 to present
- **Developer Intern**, ITAakash Strategic Software Pvt. Ltd., Mumbai — June 2026 to Aug 2026

Technical Skills

Languages / Infra: Python, SQL, TypeScript, Bash, C++, Git, Docker, FastAPI, PostgreSQL
ML / Data: PyTorch, PyTorch Geometric, NAS, Pruning, Quantization (QAT), scikit-learn, XGBoost, LightGBM, Hugging Face

Declaration: I hereby declare that the information furnished above is true to the best of my knowledge.

Place: Mumbai **Viraj Sanghavi**

Date: _____